HACKATHON SUBMISSION

ClaimFlow AI

Seven AI agents prepare the claim file. A person makes the decision.

Generative AI | Agentic AI | Multi-agent system | RAG | Human in the loop



CLAIMFLOW AI

Claim package

CF-2026-0001

FINAL: approved for processing by a person

Claimant	Ayesha Khan
Policy number	POL-1001
Incident	Collision, 28 September 2026
Claim status	Approved for processing
Package generated	2026-10-02 18:27:56 UTC
AI mode	Demo mode (scripted offline model)
Documents on file	sample_motor_policy.pdf (insurance policy), accident_report_clean.pdf (accident report), repair_quotation_clean.pdf (repair quotation), rear_bumper_damage.jpg (damage photo)

1. Decision

HUMAN DECISION	Approved for Processing Decided by Demo Reviewer on 2026-10-02 18:27:56 UTC. The reviewer's note: "Checked the findings against the cited policy pages."
AI RECOMMENDATION	Approve for Processing Advice only. It has no effect until a person decides.

The reviewer agreed with the AI recommendation.

AI explanation: The recommendation is to approve for processing. The automated checks found no blocking issue. A person makes the final decision.

Decision history

When	Reviewer	Decision	AI recommended at the time	Result	Notes
2026-10-02 18:27:56 UTC	Demo Reviewer	Approved for Processing	Approve for processing	Agreed with the AI	Checked the findings against the cited policy pages.

The AI recommends. A person decides. Nothing in this package was approved or rejected automatically.

2. The claim

Rear-ended at a traffic light on Shahrah-e-Faisal. The rear bumper and boot lid are damaged. No injuries.

3. Claim assessment

Step	Amount
Repair estimate (confirmed in a document)	185000.00
Deductible (confirmed in the policy)	25000.00
Amount after the deductible	160000.00

A real claim package, generated by the app

[Team name] [Hackathon name] [Date]

PROBLEM

A claim file is many documents that must agree

1

Is it covered?

What does the policy say about this kind of loss, the excess and the limit?

2

Is the evidence complete?

Report, quotation, photos and the policy: attached and readable?

3

Do the documents agree?

Dates, names, policy number and amounts across the form and the files.

4

How much could be paid?

The estimate minus the excess, capped at the limit.

A chat model alone is not safe here

- It can invent a policy clause
- It can miscalculate the amount
- It can accuse a claimant
- It can follow instructions hidden in an uploaded file
- It can quietly make the decision

ClaimFlow AI is built around these five risks.

EXISTING PROCESS

Today one person does every step by hand

1 Receive the claim	2 Read each document	3 Search the policy	4 Compare details	5 Do the arithmetic	6 Write up and decide
Form, policy PDF, report, quotation and photos arrive as separate files.	Long, repetitive reading before any judgement.	Finding the right clause. The excess often differs by claim type.	Dates, names and amounts checked by eye. Mismatches slip through.	Calculator or spreadsheet. One slip changes the payout.	The reasoning is rarely recorded in one place for audit.

Typical manual process, described from general domain knowledge. We did not run a time study, so this deck makes no time-saved claim.

Agents prepare the file. A person decides.

What the agents produce

- Structured claim facts from the documents
- Policy findings with page citations
- An evidence checklist and consistency checks
- Risk indicators, never judgements of honesty
- A potential amount with its working
- A recommendation with an explanation
- A claim package PDF with an audit trail

Models propose, code verifies

Every quote, date and figure a model gives is checked against the source text.

Money comes from code

The amount is a fixed formula in Decimal. A model never calculates it.

Cite it or say not found

A policy statement needs a quote found word for word on the cited page.

Two modes: Live AI uses Gemini, Groq or OpenRouter. Demo Mode runs the whole workflow offline with a scripted model, and is labelled as such.

MULTI-AGENT ARCHITECTURE

Seven agents, one shared claim state

Workflow orchestrator: runs steps in order, skips what cannot run, contains failures, saves every run

Independent steps

1 Claim intake
model proposes, code verifies

2 Policy analyst
RAG, every quote checked

3 Damage analysis
vision; limited demo in Demo Mode

Need intake

4 Evidence verification
fixed rules, no model

5 Risk analysis
fixed rules, never judges conduct

6 Claim assessment
fixed formula, no model

Needs all four

7 Compliance review

Fixed rules choose approve, request information or escalate. A model words the explanation and code checks it.

Human review

A person decides. Then the package PDF.

Shared claim state: the typed result of every step, one copy per agent. Agents never call each other.

AI WORKFLOW

One click runs seven steps and survives failures

Four outcomes per step

Done
Finished, nothing to flag.

Warning
Finished. A person should look.

Failed
Could not finish. The claim goes on.

Skipped
Something it needs is missing. Reason shown.

What happens when

No policy attached
Evidence says so. The amount is not calculated. A person gets "request more information".

A vision call fails
The damage step warns. The photos stay visible.

Intake fails
Steps that need it are skipped, each with a reason.

A quotation is attached
5 of 7 steps re-run. Policy and damage are not repeated.

Maximum model calls per claim



At most 6. Risk skips its call with no documents to compare. Damage skips with no photos.

POLICY RAG

Every finding cites its page, or says not found

1 Read

PyMuPDF, pypdf if it fails or finds no text

2 Split

By section and page. Page and section kept.

3 Embed

bge-small, or offline hashing in Demo Mode

4 Store

FAISS, or NumPy. Saved to disk.

5 Retrieve

Vector plus keyword search. The agent writes queries.

6 Verify

Quote found word for word? Amount inside it?

A verified finding

Deductible: 25,000.00

Page 3, Section 4. Excess (sample_motor_policy.pdf)

"The excess for a collision and accidental damage claim is PKR 25,000 for each claim."

When the policy is silent

Is a rental car provided while my car is repaired?

This was not found in the policy.

An answer whose quote cannot be found is dropped, not shown.

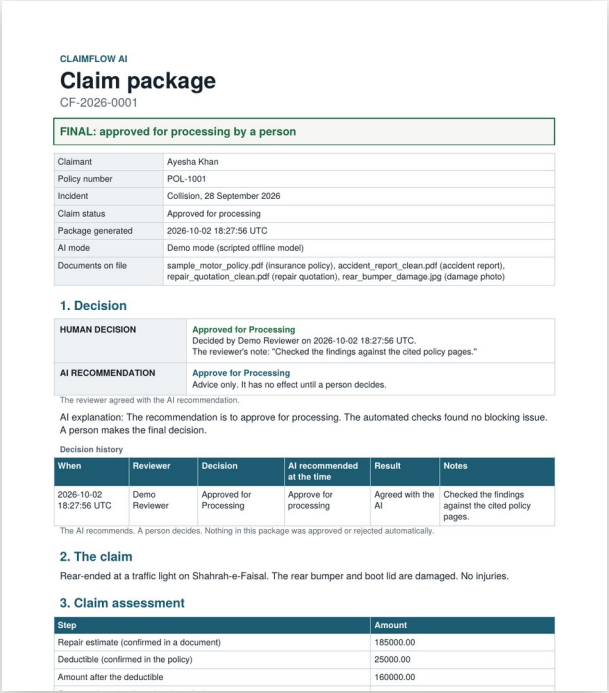
Sample policy: 6 pages, 29 chunks, 28 section labels. The index is saved and reused until the file changes.

LIVE DEMO

One clean claim, start to finish

1. Load the demo claim: four readable documents attached
2. Run seven agents: live trace, four possible outcomes
3. Policy RAG: deductible 25,000 on page 3, Section 4
4. Evidence 4 of 4 present. Risk: every planned check ran
5. Calculation in code: 185,000 minus 25,000 gives 160,000.00
6. Human review: a person approves, beside the AI recommendation
7. Claim package PDF: decision, citations, working, audit trail

Demo Mode note: the damage step is a limited demo analysis. It reads file names and basic image checks, not the photo.



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Real output: the package generated for this claim

A person decides, and the AI has guardrails

AI RECOMMENDATION

Approve for Processing (advice only)

HUMAN DECISION

Pending, until a person decides

- Only a person can approve, request information or escalate
- Overriding the AI needs a written reason
- The AI's recommendation is stored with the decision
- The package is a draft until a person decides

Responsible AI controls

- Policy quotes are checked word for word
- Amounts are calculated by code, never by a model
- Risk analysis never judges honesty. A wording guard blocks accusations
- Text inside documents is untrusted. Hidden instructions are reported, not followed
- Uploads are checked for type, size and real content
- Demo Mode and its limited photo analysis are labelled

TECHNOLOGY STACK

Plain Python, local data, swappable AI models

Interface

Streamlit, pandas. Six pages.

Agents and workflow

Python 3.10+, typed dataclass schemas, a custom orchestrator.

AI models

Gemini, Groq or OpenRouter over REST. A scripted offline model for Demo Mode.

Policy RAG

PyMuPDF (pypdf fallback), fastembed bge-small, FAISS (NumPy fallback), BM25 keyword search.

Data and documents

SQLite, files on disk, Pillow for images, ReportLab for the PDF.

Quality

338 automated tests that need no network and no API key.

Configured with environment variables. Runs locally with pip install and streamlit run app.py.

BENEFITS

Consistent, traceable and under human control

7
agents, each with a typed output

6
model calls at most, per claim

0
amounts calculated by a model

338
automated tests, no network needed

Consistent

The same checks run on every claim: evidence, consistency, risk and calculation.

Traceable

Every policy finding cites a page. Every agent run is saved, and the package carries the audit trail.

Under control

A person decides. Overrides need a reason. The AI recommendation is stored beside the decision.

Resilient

A failed step is shown, not hidden. The whole workflow also runs offline in Demo Mode.

Impact is stated as what the system does. We have not measured time saved or accuracy on real claims.

FUTURE SCOPE

What comes next, and what is limited today

Next

- OCR for scanned documents
- Percentage and additional excesses
- A check of the incident date against the policy period
- More claim types and languages
- User accounts, roles and a fuller audit log
- A persistent database for hosted use
- Evaluation on real, labelled claims
- Integration with claim systems and email

Limited today

- Scanned PDFs are refused. There is no OCR
- In Demo Mode the photo is not interpreted
- Live AI has not been run against live APIs by the team
- Percentage deductibles are flagged, not calculated
- No login: the reviewer types a name
- Only the first 60 pages of a PDF are read